

Assignment 3 — From Raw Signals to a Lean Feature Set

Feature Engineering & MLOps · Phase 3 (Time-based Features & Feature Creation) + Phase 4 (Feature Selection)

Course	Feature Engineering & MLOps
Assignment No.	3
Topic	Time-based Features, Feature Creation (Phase 3) + Filter/Wrapper/Embedded Selection (Phase 4)
Marks	____ / 100
Due Date	_____

1. Objective

This assignment deliberately spans TWO phases of the course, because in real projects they always happen back to back: first you CREATE candidate features (Phase 3), then you figure out which of them actually deserve to stay (Phase 4). You will engineer time-based and interaction features on a daily institute-performance dataset, then apply filter, wrapper, and embedded selection methods to cut a large, noisy feature set down to a lean, trustworthy one.

By the end, you should be able to answer, with evidence from your own output: "of everything I just created, what actually survives selection — and do the three selection families agree with each other?"

2. Learning Outcomes

- Engineer lag, rolling-window, cyclical, and trend features from a raw date column.
- Create an interaction feature and justify it with a measurable performance gain.
- Apply filter methods (variance threshold, correlation filtering, mutual information) to a mixed set of genuine, redundant, and noise features.
- Apply a wrapper method (RFE) and an embedded method (Lasso) to the SAME feature set, and compare what each one keeps.
- Recognize which filter techniques are NOT appropriate for a continuous target, and explain why.

3. Dataset

Use `data/raw/daily_batch_performance.csv` — 730 days of PrepEdge institute performance data.

Column	Description
date	Calendar date (2024-01-01 to 2025-12-31)

Column	Description
marketing_spend	Daily marketing spend (■)
discount_pct	Promotional discount offered that day (0–30%)
website_visits	Daily website visits
website_visits_copy	A near-duplicate of website_visits (planted on purpose)
competitor_promo_flag	1 if a competitor ran a promotion that day
is_holiday	1 if the date falls in a major holiday window
random_noise_1	Numeric column with no real relationship to the target
random_noise_2	Categorical column (A/B/C) with no real relationship to the target
new_enrollments	Target — new student enrollments that day (regression)

NOTE: Regenerate this file anytime with `python data/generate_daily_batch_data.py` — it uses a fixed random seed, so your data will always match the solution key exactly.

4. Tasks

4.1 — Phase 3: Time-based Features & Feature Creation

1. Load the dataset and sort by date. Plot `new_enrollments` over time and note, in one sentence, any pattern you can see by eye (trend, weekly rhythm, etc.).
2. **Lag features:** create `marketing_spend_lag1` and `marketing_spend_lag2`. Compute the correlation of EACH with `new_enrollments`, alongside same-day `marketing_spend`'s correlation. Report which one is strongest — you should find a lagged version beats the same-day value, and briefly explain why that makes real-world sense for a coaching institute (hint: it takes a couple of days for an ad to convert into a walk-in).
3. **Rolling features:** create a 7-day rolling mean and 7-day rolling std of `new_enrollments` itself.
4. **Cyclical encoding:** extract day-of-week and encode it with sine/cosine (`dow_sin`, `dow_cos`) rather than a plain 0–6 integer. In one sentence, explain what would go wrong if you used the plain integer instead.
5. **Trend feature:** create a simple `time_index` (0, 1, 2, ...) and fit a quick linear regression of `new_enrollments` on it alone. Report the estimated daily growth in enrollments.
6. **Interaction feature (Feature Creation):** create `spend_x_discount = marketing_spend_lag2 × discount_pct`. Using a simple linear model, compare test R^2 WITH vs WITHOUT this interaction term (keeping every other feature identical) and report whether it helps.
7. Drop the leading rows that now contain NaN (from the lag/rolling features) before moving on to Section 4.2 — explain in one sentence why these rows cannot be mean/median-imputed the way an ordinary missing value would be.

4.2 — Phase 4: Feature Selection

You should now have a feature set that includes: the original columns, your engineered lag/rolling/cyclical/trend/interaction features, the planted redundant column (`website_visits_copy`), and the two noise columns. Apply each family of selection method to this full set.

1. **Variance threshold:** check whether any feature is near-constant. (You may find none are — if so, say so; don't force a removal that isn't justified by the data.)
2. **Correlation filtering:** build a correlation heatmap of all numeric features. Identify the pair you'd expect to be flagged, confirm it in your heatmap, and decide which one of the pair to drop.
3. **Mutual information:** compute `mutual_info_regression` for every numeric feature against `new_enrollments`. Rank the features and report where the two noise columns land.
4. **Trick question — read carefully:** the Filter Methods topic also covers chi-square and ANOVA F-test. Explain, in 2–3 sentences, why NEITHER is the right tool for filtering features against `new_enrollments` in this assignment (hint: think about what type of TARGET each test expects).
5. **Wrapper method (RFE):** run Recursive Feature Elimination with a linear regression model to select the top 6 features. Report the ranking.
6. **Embedded method (Lasso):** fit LassoCV on the same (standardized) feature set. Report which coefficients shrink to exactly zero, or nearly zero.

7. Compare all three: build one small table — rows = every candidate feature, columns = filter / wrapper / embedded, cells = kept or dropped by that method. Do the two noise columns get dropped by ALL THREE approaches? Does the redundant `website_visits_copy` get caught by all three, or only some?

4.3 — Reflection questions (short answer)

1. Which single engineered feature from Section 4.1 turned out to be most useful according to your Section 4.2 analysis? Does that match your intuition from when you first created it?
2. If a wrapper method and an embedded method DISAGREE on whether to keep a feature, which would you trust more for this dataset, and why?
3. Suppose next month's data included a BRAND NEW competitor promotion column you'd never seen before. Would your selection results from this assignment still apply, or would you need to redo the analysis? Justify your answer.

5. Submission Instructions

1. In your existing course GitHub repository, use (or create) the top-level `Assignments` folder.
2. Add your completed notebook as `Assignment3_TimeFeatures_Selection.ipynb`. Run every cell top-to-bottom with no errors, and keep outputs visible before committing.
3. Structure the notebook with clear markdown headers matching Section 4.1, 4.2, and 4.3 above.
4. Commit and push your changes:

```
git add Assignments/Assignment3_TimeFeatures_Selection.ipynb
git commit -m "Assignment 3: Time-based features + feature selection"
git push
```

Finally, share your GitHub repository link with the instructor by the due date above. Grading is done directly from the `Assignments` folder — make sure `Assignments 1`, `2`, and `3` are all there and pushed before the deadline.

IMPORTANT: Only the notebook committed to your `Assignments` folder on GitHub will be evaluated. Local-only or emailed notebooks will not be graded.

6. Grading Rubric

Criteria	Weight
Phase 3: lag/rolling/cyclical/trend features correctly implemented	20%
Phase 3: interaction feature created and its benefit measured (not just asserted)	15%
Phase 4: all three filter techniques correctly applied and interpreted	20%
Phase 4: RFE and Lasso correctly applied, with a clear kept-vs-dropped comparison table	20%
Correct reasoning on the chi-square/ANOVA trick question	10%
Reflection questions answered thoughtfully; code quality and markdown structure	15%

7. Bonus (Optional, +10% extra credit)

Add a Random Forest embedded-importance ranking alongside Lasso's. Does the tree-based ranking agree with Lasso on which of the two near-duplicate columns (`website_visits` vs `website_visits_copy`) to keep? If they disagree, explain why, referencing how each model type handles correlated features differently.